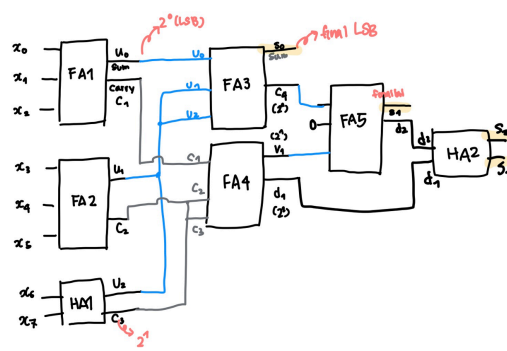


Exercise 02 - Written Part

01. Given 8 binary inputs $x_0, \dots, x_7$ and 4 binary outputs $s_0, \dots, s_3$ design a circuit where the output is the binary representation of the count of the 1 values in the input, i.e.

$$\sum_{i=0}^7 x_i = \sum_{i=0}^3 2^i s_i$$



02. Let $x: i32$, when $x \& (x-1)$ is true, why?

- If x is a power of 2, the result is 0 $\rightarrow$ false.
- If x is not a power of 2, the result is nonzero $\rightarrow$ true.

Example

Power of 2

$$x = 8 = 1000_2 \quad x-1 = 7 = 0111_2$$

They are opposite so if you $\&$ operate this, this becomes 0 (false)

Not power of 2

$$x = 2 = 10_2 \quad x-1 = 1 = 01_2$$

After and operation, this becomes 0 (false)

If x is not a power of 2, then x has more than one 1-bit. In that case, at least one 1-bit remains after $x \& (x-1)$, so the result is always nonzero (true).

03. The following are binary values of some precision (32 bit) IEEE754 floating point values:

3FCF 1BBD
FF80 0000
8800 0000
3E4C CCCD
3FDD B3D7

Which numbers do they represent exactly in decimal and which real number are they supposed to represent?

Example: C000 000 = exact decimal -2.0 real -2.0

Approach

The IEEE 754 standard specifies a binary32 as having:

- Sign bit: 1 bit
- Exponent width: 8 bits
- Significand precision: 24 bits (23 explicitly stored)

$$N = (-1)^s \times (1.f) \times 2^{e-127}$$

- [IEEE Reference](#)
- [Equation Reference](#)

Hex to 32bit Binary

Example : 3FCF1BBD -> 00111111 11001111 00011011 10111101

Sign, Exponent, Fraction

- sign = 1 bit (0 - positive, 1 - negative)
- exponent = 8 bits
- fraction = last 23 bits

Calculate Exponent

Exponent Field : Decimal - bias 127 (we need bias for the negative values)

Calculate Mantissa

 Mantissa = 1 + fraction(23 bits part)

$$\text{Fraction} = b_1 \times 2^{-1} + b_2 \times 2^{-2} \dots$$

Final Value

$$N = (-1)^s \times (1.f) \times 2^{e-127}$$

Hex: 3FCF1BBD

1. Binary: 00111111 11001111 00011011 10111101
2. Sign/Exponent/Fraction
 - Sign bit: 0 (positive)
 - Exponent field (raw 8 bits): 01111111 (decimal 127)
 - Fraction bits (23 bits): 1001111000110111011101
4. Actual exponent ($E = \text{exponent_field} - \text{bias}$): 0
5. Mantissa = $1 + \text{fraction}$ ($\rightarrow \text{fraction} = \sum b_i \cdot 2^{-i}$) Mantissa exact (as fraction) = $13573053/2^{23} = 13573053 / 8388608$

Approximate decimal (f32, 17 significant digits) =
1.61803400516510010

Hex: FF800000

1. Binary: 11111111 10000000 00000000 00000000
2. Sign/Exponent/Fraction
 - Sign bit: 1 (negative)
 - Exponent field (raw 8 bits): 11111111 (decimal 255)
 - fraction = 0, Classification: -infinity

Negative Infinity

Hex: 88000000

1. Binary: 10001000 00000000 00000000 00000000
2. Sign/Exponent/Fraction
 - Sign bit: 1 (negative)
 - Exponent field (raw 8 bits): 00010000 (decimal 16)
 - Fraction bits (23 bits): 00000000000000000000000
3. Actual exponent ($E = \text{exponent_field} - \text{bias}$): -111
4. Mantissa = $1 + \text{fraction}$ ($\rightarrow \text{fraction} = \sum b_i \cdot 2^{-i}$) Mantissa exact (as fraction) = $8388608/2^{23} = 8388608 / 8388608$

